

Picas y Fijas

Practical strategy guide for codes with 3 to 6 positions

Core idea: a strong guess is not merely one that looks close to the code. It is the guess that eliminates the most possibilities regardless of the answer.

What this guide covers

Strategies for numbers and colors, with and without repeats, recommended openings, clue interpretation, and a systematic method for avoiding contradictions. These are practical heuristics rather than a proof of the absolute minimum number of turns.

Golden rule

After every guess, keep only the codes that would produce exactly the same number of bulls (F) and cows (P) if they were the secret. Choose the next guess to split that remaining set into small groups.

Concept	Strategic meaning
Bull / F	Correct symbol in the correct position.
Cow / P	Correct symbol in a different position.
0F 0P	Discard every symbol in that guess.
F + P	Total number of symbols from the guess that occur in the secret.

1. Universal elimination method

1. Explore

Without repeats, open with distinct symbols and cover new symbols. With repeats, include a duplicate early to measure multiplicity.

2. Record

Write down the exact F and P result. F+P alone is not enough when position matters.

3. Filter

Remove every candidate that fails to reproduce every previous clue.

4. Split

If many candidates remain, use a probe that compares positions or introduces new symbols. A probe does not need to be a possible winner.

5. Solve

When one or very few candidates remain, play a compatible candidate or the guess that separates them best.

How to read a clue

Answer	Next action
0F 0P	Replace all symbols with new ones.
0F and several P	The symbols occur, but none is in that position. Reorder them methodically.
Several F	Provisionally preserve those positions and change the others to confirm them.
F+P below code length	Some symbols are missing; introduce new ones without losing all positional information.

Do not change every symbol and every position at once after finding matches. Change one controlled element so the new clue remains interpretable.

About 123 - 456 - 789

This is a valid coverage strategy for three digits without repeats. F+P tells how many digits from each block belong to the secret. However, it can spend three turns before studying positions, initially omits 0, and may be inefficient with a six-guess limit. Stop as soon as three digits have been located. If 123 gives two matches and 456 gives one, all three digits are already known: do not play 789.

2. Strategies without repeats

First identify the set of symbols, then determine their positions. Stop introducing symbols once F+P evidence accounts for the whole code.

Positions	Suggested opening	Practical plan
3	123; then 456 if needed	Count membership with F+P. Stop at 3 symbols, remember 0, then swap two positions at a time.
4	0123	If F+P=4, only order remains. Otherwise replace two symbols with 45, then complete with 67/89 if needed.
5	01234	With high F+P, preserve the block and test controlled permutations. With low F+P, replace 2 or 3 symbols with 567.
6	012345	Replace groups of two with 67 and then 89. Once six symbols are known, resolve order through swaps.

Swap technique

Once the symbols are known, swap only two positions. If F rises by two, both positions were corrected; if it falls by two, both were correct before. If it is unchanged, combine that fact with another controlled swap.

Example: 3 digits without repeats

Turn	Guess	Clue	Deduction
1	123	0F 2P	Two of {1,2,3} occur, both misplaced.
2	456	0F 1P	One of {4,5,6} occurs. All three symbols are known; skip 789.
3	214	1F 1P	Compare positions of 1 and 2 and test whether 4 is the third symbol.
4+	Compatible candidate	-	Only play codes satisfying every clue.

With a short guess limit, the second play must depend on the first clue. A rigid sequence wastes information.

3. Strategies with repeats

You must discover both symbol identity and multiplicity. All-distinct openings are weaker at detecting duplicates.

Positions	Suggested opening	Purpose
3	112	Detect whether 1 repeats while collecting positional information.
4	1122	Measure the contribution and multiplicity of two symbols.
5	11223	Test two pairs plus one additional symbol.
6	112233	Test three pairs, then replace absent pairs with 44/55/66.

Multiplicity test

Playing one symbol in every position (for example 1111) makes $F+P$ equal its exact multiplicity. This is clear but expensive, so reserve it for likely duplicates or candidates differing only in counts.

With repeats, $F+P=2$ does not necessarily mean two different symbols; it may be one symbol appearing twice.

4. Color games

The logic is identical. Treat colors as symbols and account for the actual palette size (4, 6, or 8). Smaller palettes make duplicate testing more important.

Palette	Without repeats	With repeats
4 colors	A distinct-color opening covers almost the entire universe for 3 or 4 positions.	For 4 positions use AABB, then CCDD. $F+P$ measures each pair.
6 colors	Cover as many different colors as positions; add the remainder only if $F+P$ is incomplete.	Use pairs such as AABB/CDDD or a mixed AABC opening.
8 colors	Explore in blocks and stop when total matches fill the code.	Balance new colors with duplicates; do not test all eight unnecessarily.

5. Strategy by guess limit

Setting	Priority
Unlimited	Maximize certainty; pure presence or multiplicity probes are affordable.
10 guesses	Balance: 1-3 exploration turns, then compatible candidates.
6 guesses	Every guess should explore while moving toward a solution. Avoid automatic full-alphabet coverage.
Timer	Keep a simple guess/F/P table. A perfect strategy that takes too long loses value.

Quick decision tree

After a guess	Decision
$F+P = \text{code length}$	Introduce no new symbols; solve order and, where applicable, counts.
$F+P = 0$	Replace all symbols.
$0 < F+P < \text{length}$	Keep part of the guess and introduce enough new symbols to complete the code.
2-5 candidates remain	Choose a guess producing different answers across them, even if it is not a candidate.
1 candidate remains	Play it.

Common mistakes

- Repeating a guess that cannot separate candidates.
- Ignoring 0 in number mode.
- Changing too many positions after several F.
- Playing a code that contradicts an earlier clue.
- Treating $F+P$ as a count of distinct symbols when repeats are allowed.
- Following a fixed sequence after all symbols are already located.

Tracking sheet

Turn	Guess	F	P	F+P	Deduction / remaining candidates
1					
2					
3					
4					
5					

Conclusion: strategies do exist. The strongest practical method treats each clue as a mathematical constraint, removes incompatible candidates, and chooses the next guess for the information it can produce.